

Microsoft Azure

Azure Data Factory - Control Flow Activity Project

Aishwarya Mohan

Introduction

As we have done earlier (Resource Group, Storage Account, Data Factory creation, Source and Destination containers, and the Copy Activity foundation), we now continue with Control Flow activities inside Azure Data Factory Studio.

This project covers the following Control Flow activities and triggers:

- Copy Data activity with column mapping
- Delete activity with logging settings
- Triggers: Manual, Schedule, Tumbling Window, and Storage Events
- Set Variable – area calculation (fixed values and parameterised runtime input)
- Append Variable
- Get Metadata
- Execute Pipeline

All activities were configured and executed on the Microsoft Azure platform.
The screenshots below show the exact configuration and successful runs.

Step 1: Copy Data Activity with Mapping

Open pipeline1. Add Copy data activity (Source to Destination).

Source dataset: SD_01 (Pokemon.csv). Sink dataset: DS_01.

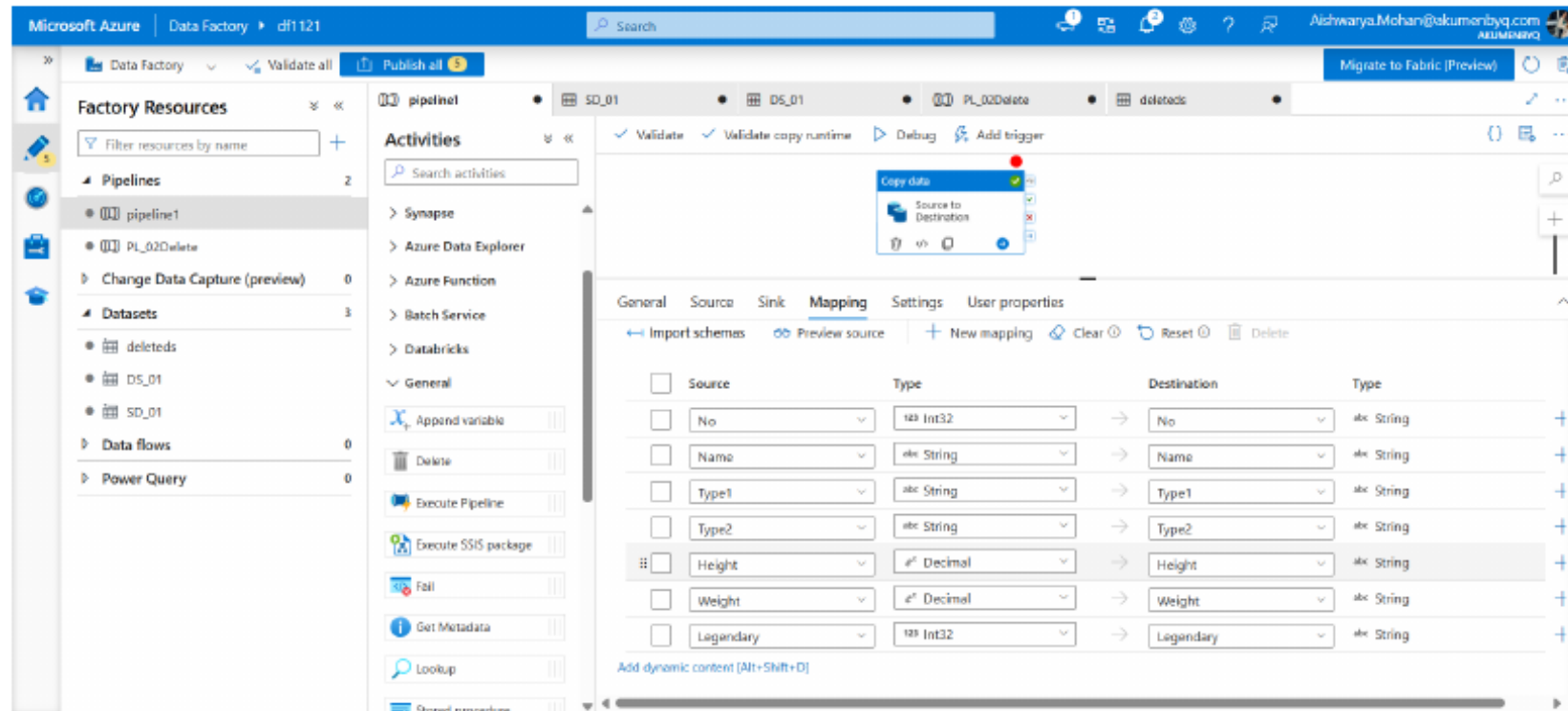
Mapping tab – map columns: No, Name, Type1, Type2, Height, Weight, Legendary.

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' and 'Datasets' sections. 'pipeline1' is selected under Pipelines, and 'SD_01' is selected under Datasets. The main workspace shows the 'Activities' list on the left and the 'Copy data' activity configuration on the right. The 'Copy data' activity is currently in the 'Source' tab, showing the 'Source dataset' as 'SD_01'. The 'File path type' is set to 'File path in dataset'. The 'Filter by last modified' and 'Enable partitions discovery' options are visible. The 'Mapping' tab is also visible, showing the column mapping for the activity.

Step 1 (continued): Column Mapping Configuration

Import schemas and map source columns to destination columns.

Type conversions are applied as shown (Int32 → String, Decimal → String, etc.).



The screenshot displays the Microsoft Azure Data Factory interface. The top navigation bar shows 'Microsoft Azure | Data Factory | df1121'. The left sidebar contains 'Factory Resources' with sections for Pipelines (2 items), Datasets (3 items), Data flows (0 items), and Power Query (0 items). The central pane shows 'Activities' with a search bar and a list of activity types: Synapse, Azure Data Explorer, Azure Function, Batch Service, and Databricks. The main workspace shows a 'Copy data' activity with a 'Mapping' tab selected. The mapping table shows source columns being mapped to destination columns with type conversions.

Source	Type	Destination	Type
No	Int32	No	String
Name	String	Name	String
Type1	String	Type1	String
Type2	String	Type2	String
Height	Decimal	Height	String
Weight	Decimal	Weight	String
Legendary	Int32	Legendary	String

Step 2: Successful Copy Data Pipeline Run

Debug or Trigger the pipeline. Pipeline status: Succeeded.

Activity 'Source to Destination' completed successfully. File appears in destination container.

The screenshot displays the Microsoft Azure Data Factory console. The top navigation bar shows 'Microsoft Azure | Data Factory | dl1121'. The left sidebar contains 'Factory Resources' with a search bar and a list of resources: Pipelines (2), Datasets (3), Data flows (0), and Power Query (0). The 'Pipelines' section is expanded, showing 'pipeline1' and 'PL_02Delete'. The 'Activities' section on the right shows a search bar and a list of activities: Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, and General. The 'General' section is expanded, showing 'Append variable', 'Delete', 'Execute Pipeline', 'Execute SSIS package', 'Fail', 'Get Metadata', 'Lookup', and 'Stored procedure'. The main canvas shows a pipeline diagram with a 'Copy data' activity. Below the canvas, the 'Output' tab is selected, showing the 'Pipeline run' details. The pipeline run ID is '6d8b7891-789e-4e87-a387-ef1e21b1178f'. The pipeline status is 'Succeeded'. The table below shows the activity details:

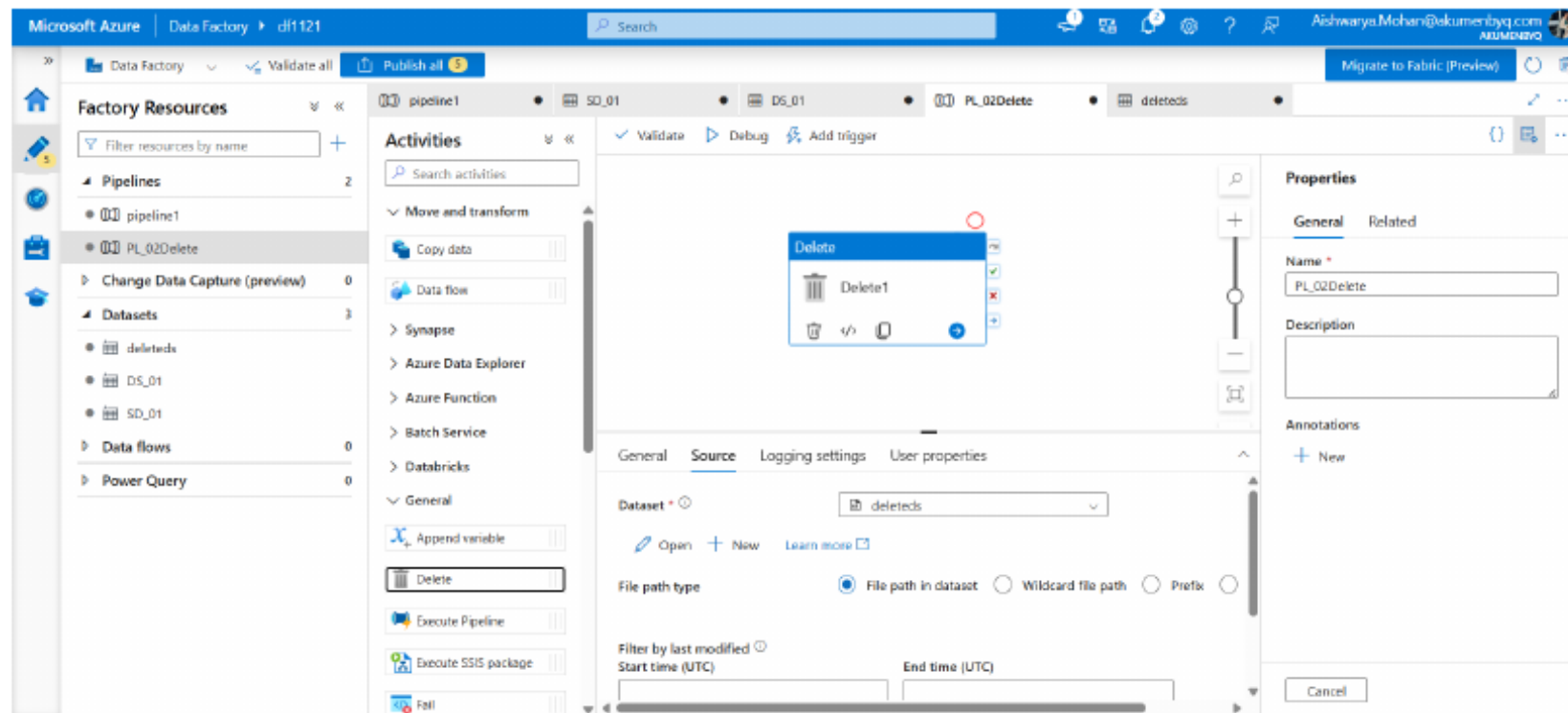
Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Source to Destination	Succeeded	Copy data	8/14/2026, 12:53:30 PM	17s	AutoResolveIntegrationRuntime (East US)

Step 3: Delete Activity

Create pipeline PL_02Delete (or add Delete activity).

Configure Delete activity with dataset 'deletedes' pointing to ABSTRACT.docx.

File path type: File path in dataset.



Step 3 (continued): Delete Activity Successful Run

Pipeline status: Succeeded. The target file is deleted from the container.

The screenshot displays the Microsoft Azure Data Factory interface. The top navigation bar shows 'Microsoft Azure | Data Factory | df1121'. The left sidebar lists 'Factory Resources' including Pipelines (2), Datasets (3), Data flows (0), and Power Query (0). The 'Pipelines' section is expanded, showing 'pipeline1' and 'PL_Q2Delete'. The 'Activities' section is also expanded, showing 'Delete' and 'Delete1'. The main canvas shows a 'Delete' activity with a green checkmark, indicating success. The 'Output' tab is selected, showing the pipeline status as 'Succeeded' with a green checkmark. The pipeline ID is '79095caf-a679-4701-85b6-d0acc55f2d4c'. The activity name is 'Delete1', and the status is 'Succeeded'. The run start time is '8/14/2026, 12:57:21 PM'. The 'Properties' panel on the right shows the 'General' tab with the name 'PL_Q2Delete' and a description field. The 'Annotations' section has a '+ New' button. A 'Cancel' button is visible at the bottom right.

Microsoft Azure | Data Factory | df1121

Search

Validate all Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines 2

- pipeline1
- PL_Q2Delete

Change Data Capture (preview) 0

Datasets 3

- deleted1
- DS_01
- SD_01

Data flows 0

Power Query 0

Activities

Search activities

Move and transform

- Copy data
- Data flow

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

- Append variable
- Delete
- Execute Pipeline
- Execute SSIS package
- Fail

Validate Debug Add trigger

Delete

Delete1

Parameters Variables Settings Output

Pipeline 79095caf-a679-4701-85b6-d0acc55f2d4c

run ID 5b6-d0acc55f2d4c

Pipeline status Succeeded

View debug run consumption

All status

Monitor in Azure Metrics Export to CSV

Showing 1 - 1 of 1 items

Activity name	Activity st...	Activit...	Run start
Delete1	Succeeded	Delete	8/14/2026, 12:57:21 PM

Properties

General Related

Name PL_Q2Delete

Description

Annotations

+ New

Cancel

Step 4: Delete Activity – Logging Settings

Enable logging on the Delete activity.

Select logging account linked service and set Folder path (e.g. source).

This keeps a log of deleted files for audit purposes.

The screenshot displays the Microsoft Azure Data Factory portal. The top navigation bar shows 'Data Factory' and 'df1121'. The left sidebar contains 'Factory Resources' with a search bar and a list of resources: Pipelines (2), Datasets (3), Data flows (0), and Power Query (0). The 'Pipelines' section is expanded, showing 'pipeline1' and 'PL_02Delete'. The 'Activities' section is also expanded, showing a list of activity types: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, General, HDInsight, Iteration & conditionals, Machine Learning, and Power Query. The 'Delete' activity is selected, and its properties are shown in the main pane. The 'Logging settings' tab is active, showing the following configuration:

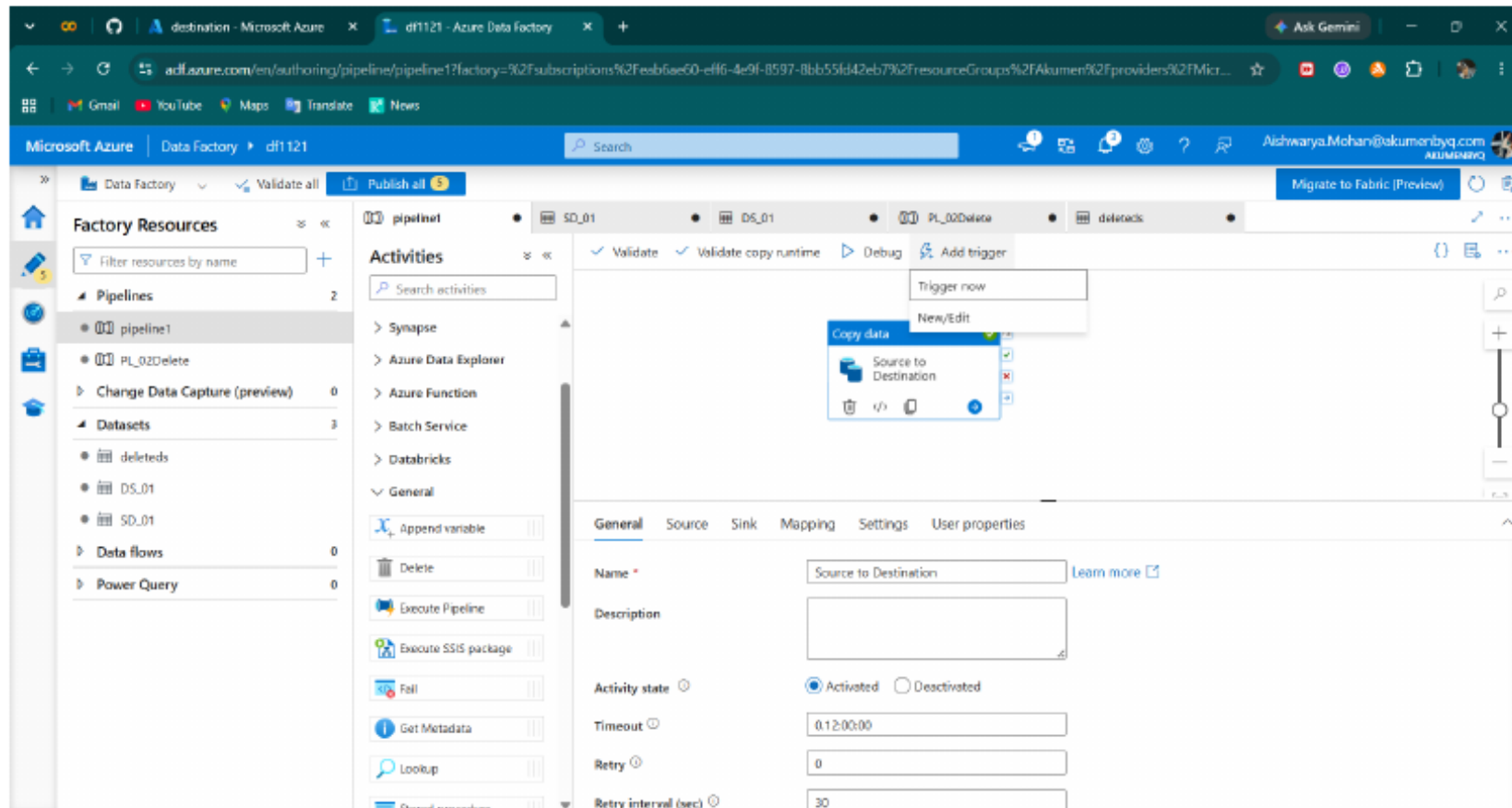
- Enable logging: ☒
- Logging settings: ☐
- Logging account linked service: (with 'Test connection', 'Edit', and 'New' buttons)
- Folder path: (with a 'Browse' button)

The 'Delete' activity is also visible in the 'Activities' pane, showing a trash icon and the name 'Delete1'.

Step 5: Manual Trigger (Trigger now / Debug)

Use Debug or 'Trigger now' to execute the pipeline manually.

View successful pipeline run and activity output in the monitoring pane.



Step 6: Schedule Trigger

Create a new Schedule trigger (trigger1).

Set Start date/time, Time zone (e.g. Chennai, Kolkata, Mumbai, New Delhi), and Recurrence (e.g. every 15 minutes).

Publish all changes and start the trigger.

The screenshot displays the Microsoft Azure Data Factory portal interface. On the left, the 'Factory Resources' pane shows a hierarchy: Pipelines (1), Datasets (2), Data flows (0), and Power Query (0). The 'Activities' pane lists various services like Move and transform, Synapse, Azure Data Explorer, etc. The main workspace shows a pipeline named 'pipeline1' with two datasets, 'DelimitedTest1' and 'DelimitedTest2'. A 'Copy data' activity is visible in the workspace. On the right, the 'New trigger' dialog box is open. It contains the following fields and options:

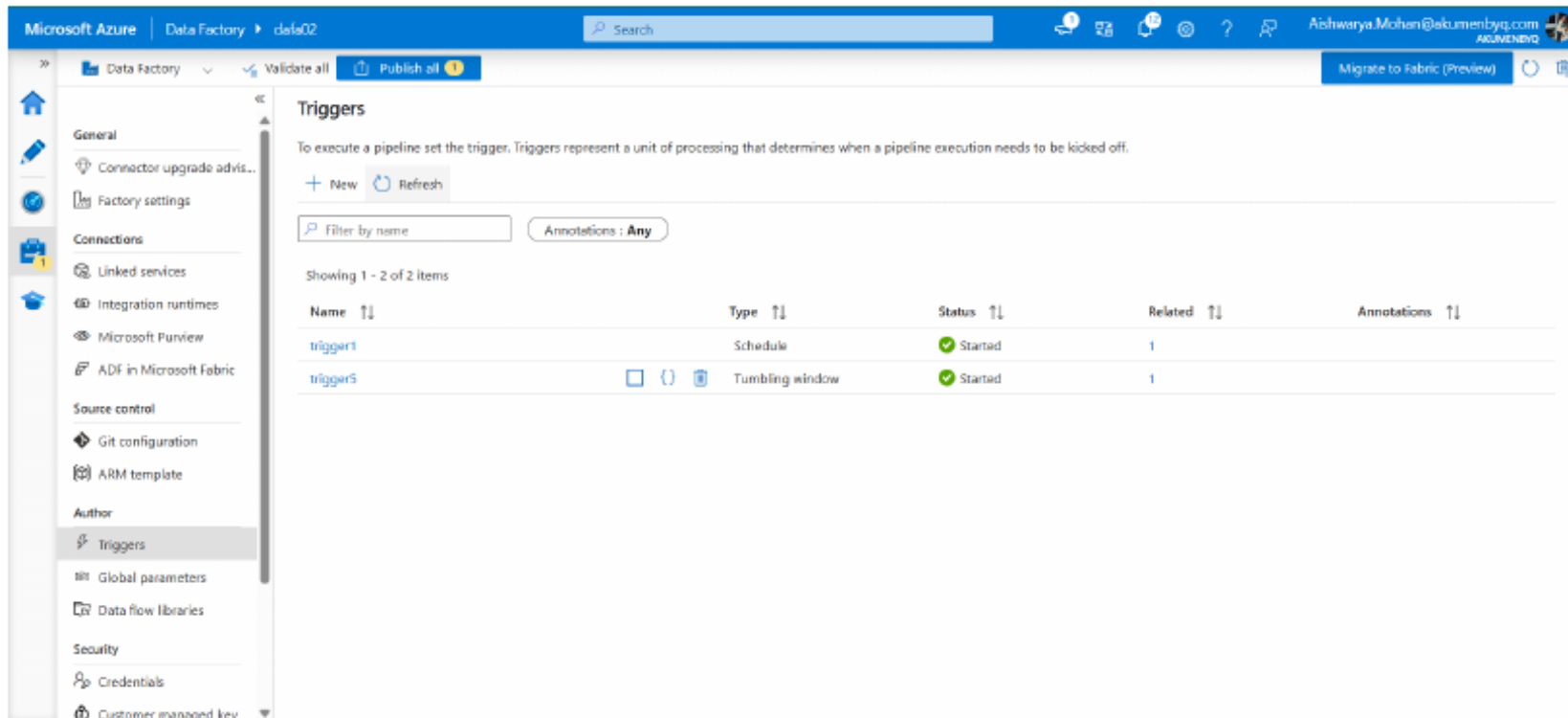
- Name:** trigger1
- Description:** (empty text box)
- Type:** Schedule (selected from a dropdown)
- Start date:** 8/17/2026, 9:24:33 PM
- Time zone:** Chennai, Kolkata, Mumbai, New Delhi (UTC+5:30) (selected from a dropdown)
- Recurrence:** Every 15 Minute(s) (selected from a dropdown)
- Specify an end date:** (unchecked checkbox)
- Annotations:** + New
- Start trigger:** (checked checkbox with label 'Start trigger on creation')

At the bottom of the dialog are 'OK' and 'Cancel' buttons.

Step 7: Tumbling Window Trigger

Create a Tumbling window trigger.

Configure the window size and start it. Status appears as 'Started' in the Triggers list.



The screenshot shows the Microsoft Azure Data Factory interface. The left sidebar contains navigation options: General, Connections, Source control, Author, and Security. The 'Triggers' option under the 'Author' section is selected. The main panel displays the 'Triggers' page with a description: 'To execute a pipeline set the trigger. Triggers represent a unit of processing that determines when a pipeline execution needs to be kicked off.' Below this, there are '+ New' and 'Refresh' buttons. A search bar labeled 'Filter by name' and a dropdown for 'Annotations: Any' are present. The table shows 'Showing 1 - 2 of 2 items'.

Name	Type	Status	Related	Annotations
trigger1	Schedule	Started	1	
trigger5	Tumbling window	Started	1	

Step 8: Storage Event Trigger

Create a Storage events trigger (Blob created).

Select Azure subscription, Storage account, and Container (source).

Publish and start. Upload a file to the source container to fire the trigger.

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a pipeline named 'pipeline1' and two datasets, 'DelimitedText1' and 'DelimitedText2'. The 'Activities' pane shows a list of activities including 'Move and transform', 'Synapse', 'Azure Data Explorer', 'Azure Function', 'Batch Service', 'Databricks', 'General', 'HDInsight', 'Iteration & conditionals', 'Machine Learning', and 'Power Query'. The 'Output' pane shows the 'Pipeline run ID' as '719c5db9-d757-4f4d-8c31-7228a785ae4f' and the 'Activity name' as 'Copy data1', which has a status of 'Succeeded'.

The 'Edit trigger' pane on the right is configured for a 'BlobEventsTrigger'. The 'Name' is 'trigger1'. The 'Type' is 'BlobEventsTrigger'. The 'Account selection method' is 'From Azure subscription'. The 'Azure subscription' is 'Azure subscription 1 (eab6ae60-ef6-4e9f-8597-8bb556d42eb7)'. The 'Storage account name' is 'saveacco'. The 'Container name' is 'source'. The 'Blob path begins with' field is empty. The 'Continue' button is highlighted.

Step 9: Set Variable – Area Calculation (Fixed Values)

Create pipeline2. Add three Set variable activities: length, breadth, area.

Set length = 10, breadth = 10.

For area use the expression: `@mul(variables('length'), variables('breadth'))`

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'pipeline2' selected under 'Pipelines'. The main workspace shows the 'pipeline2' canvas with three 'Set variable' activities arranged vertically. The top activity is 'length', the middle is 'breadth', and the bottom is 'area'. The 'area' activity is currently selected, and its properties are shown in the right-hand pane. The 'General' tab of the properties pane shows the variable name 'length', a description field, and the activity state set to 'Activated'. The 'Settings' and 'User properties' tabs are also visible. The top navigation bar includes options like 'Validate', 'Debug', and 'Add trigger'.

Step 9 (continued): Setting Fixed Values and Expression

length variable is set to the value 10.

area variable uses the multiply expression on length and breadth variables.

The screenshot displays the Microsoft Azure Data Factory portal. The left sidebar shows the 'Factory Resources' tree with 'pipeline2' selected under 'Control flow activity'. The main canvas shows the 'Activities' list on the left and the 'pipeline2' canvas in the center. Three 'Set variable' activities are visible: 'length', 'breadth', and 'area'. The 'length' activity is highlighted, and its 'Settings' tab is active in the right-hand 'Properties' pane. In the 'Settings' tab, the 'Variable type' is set to 'Pipeline variable', the 'Name' is 'length', and the 'Value' is '10'. The 'Value' field has a green 'G' icon next to it, indicating a dynamic content placeholder. The 'Properties' pane also shows the 'General' tab with the 'Name' set to 'pipeline2' and a 'Description' field. The 'Annotations' section at the bottom right has a '+ New' button and a 'Cancel' button.

Step 9 (continued): Expression Builder for Area

Pipeline expression builder: @mul(variables('length'), variables('breadth'))

This calculates area = length × breadth.

The screenshot displays the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane shows a list of pipelines, with 'pipeline2' selected. The main workspace shows the 'Activities' pane for 'pipeline2', which includes a 'Set variable' activity for 'length' and another for 'breadth'. The 'Settings' tab for the 'Set variable' activity is active, showing the 'Variable type' as 'Pipeline variable' and the 'Name' as 'area'. The 'Value' field is empty, and the 'Add dynamic content' button is visible. On the right, the 'Pipeline expression builder' dialog is open, showing the expression '@mul(variables('length'),variables('breadth'))' in the text area. Below the text area, the 'Variables' tab is selected, showing a list of variables: 'area', 'breadth', and 'length'. The 'length' variable is highlighted, and a 'click to insert expression' button is visible next to it. The 'OK' and 'Cancel' buttons are at the bottom of the dialog.

Microsoft Azure | Data Factory | df1121

Explore upgrade options to Microsoft Fabric Data Factory. Get started with guided support. [Learn more](#)

Factory Resources

- Pipelines
- pipeline1
- Control flow activity
- pipeline2
- Change Data Capture (preview)
- Datasets
- Data flows
- Power Query

Activities

- Execute Pipeline
- Execute SSIS package
- Fail
- Get Metadata
- Lookup
- Stored procedure
- Script
- Set variable
- Validation
- Web
- WebHook
- Wait

pipeline2

Set variable

(x) length

Set variable

(x) breadth

Set variable

(x) area

General Settings User properties

Variable type Pipeline variable

Name *

area

Value *

Add dynamic content

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

@mul(variables('length'),variables('breadth'))

Clear contents

Parameters System variables Functions Variables

Search

area

breadth

length

click to insert expression

OK Cancel

Step 9 (continued): Successful Run – Area = 100

All three Set variable activities succeeded.

With length=10 and breadth=10 the calculated area is 100.

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'pipeline2' selected. The main canvas shows the pipeline design with three 'Set variable' activities: 'length', 'breadth', and 'area'. The 'Output' tab at the bottom shows a table of activity results.

Activity name	Activity status	Activity...	Run start
area	✓ Succeeded	Set variable	8/17/2026, 2:28:02 PM
breadth	✓ Succeeded	Set variable	8/17/2026, 2:28:02 PM
length	✓ Succeeded	Set variable	8/17/2026, 2:28:02 PM

Step 10: Set Variable – Parameterised Area Calculation

Add pipeline parameters: length, breadth (and optionally area).

Change Set variable activities to use @pipeline().parameters.length and @pipeline().parameters.breadth. Supply values at pipeline run time.

The screenshot displays the Microsoft Azure Data Factory Studio interface. The top navigation bar shows 'Microsoft Azure | Data Factory | df01121'. The left sidebar contains 'Factory Resources' with a search bar and a list of resources: Pipelines (1), control flow activity (1), set variable (1), Change Data Capture (preview) (0), Datasets (0), Data flows (0), and Power Query (0). The 'Activities' pane on the left lists various activities, including 'Set variable'. The main canvas shows three 'Set variable' activities arranged vertically, each with a variable name in parentheses: 'length', 'area', and 'breadth'. The 'Parameters' tab is selected, showing a table of parameters:

Name	Type	Default value
length	Int	Value
area	Int	Value
breadth	Int	Value

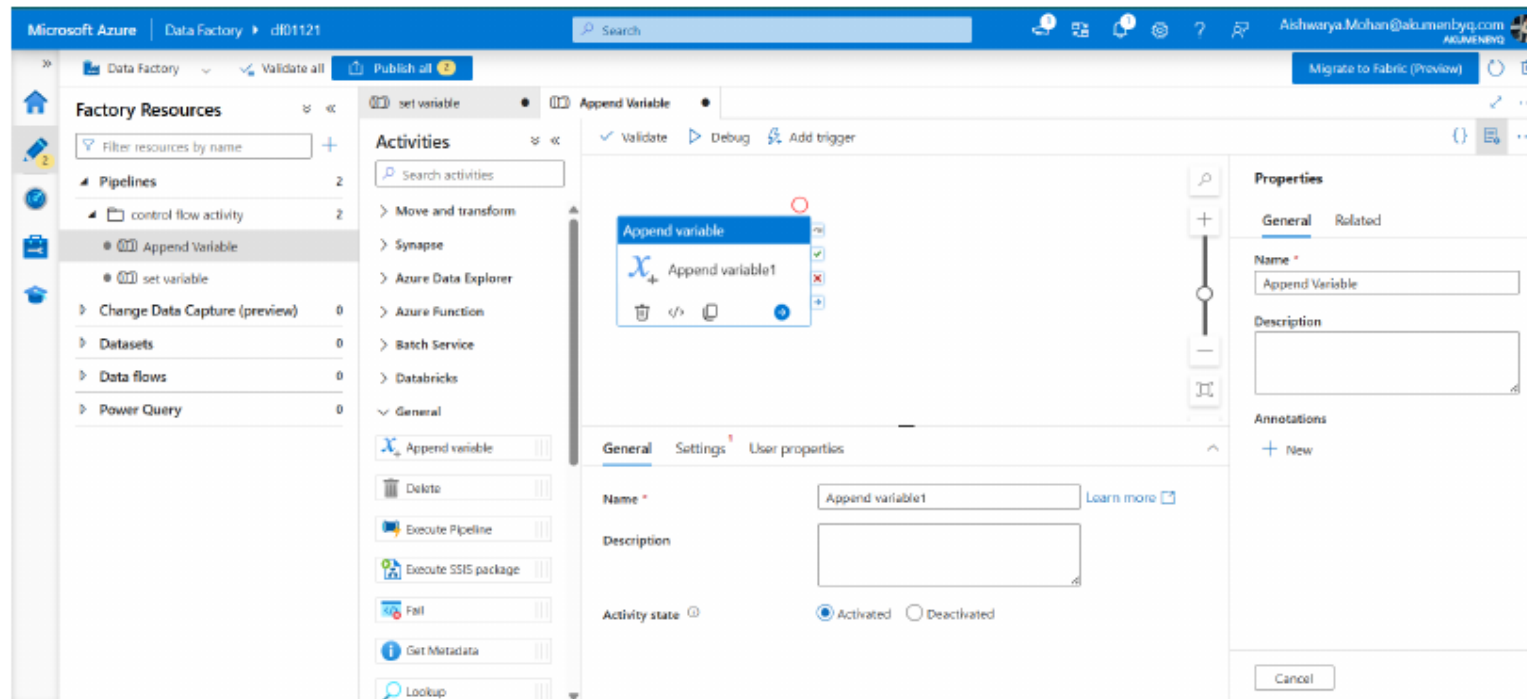
The right sidebar shows the 'Properties' pane with the 'General' tab selected. It includes fields for 'Name' (set variable), 'Description', and 'Annotations'.

Step 11: Append Variable Activity

Create a new pipeline (Append Variable).

Declare an Array variable (e.g. marks) with initial values.

Add Append variable activity to append a new value, then pass output to Set variable to view it.



Step 11 (continued): Append Variable – Array Result

After appending, the array contains the original values plus the new value.
The final array is visible in the Set variable output.

The screenshot shows the Microsoft Azure Data Factory interface. The 'Factory Resources' pane on the left lists various activities, with 'Append Variable' selected under the 'Control flow activity' category. The main canvas displays the 'Append Variable' activity configuration. The 'Variables' tab is active, showing a table of variables to be appended. The table has columns for 'Name', 'Type', and 'Default value'. Two variables are listed: 'marks' and 'output', both of type 'Array'. The 'marks' variable has a default value of '["30","45","29"]' and the 'output' variable has a default value of '["48"]'. The 'Properties' pane on the right shows the 'General' tab with the 'Name' field set to 'Append Variable' and a 'Description' field. The 'Annotations' section has a '+ New' button. At the bottom right, there are 'Cancel' and 'Show desktop' buttons.

Name	Type	Default value
marks	Array	["30","45","29"]
output	Array	["48"]

Step 11 (continued): Viewing Append Variable Output via Set Variable

Connect Append variable → Set variable.

The output shows the complete array after the append operation.

The screenshot displays the Microsoft Azure Data Factory interface. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, and 'Append Variable' selected. The main canvas shows a workflow with two activities: 'Append variable' (labeled 'Append variable1') and 'Set variable' (labeled 'Set variable1'). A green arrow connects the output of 'Append variable' to the input of 'Set variable'. The 'Set variable' activity is selected, and its 'Output' pane is open, showing a JSON object:

```
{  "name": "output",  "value": [    "30",    "45",    "29",    "90"  ]}
```

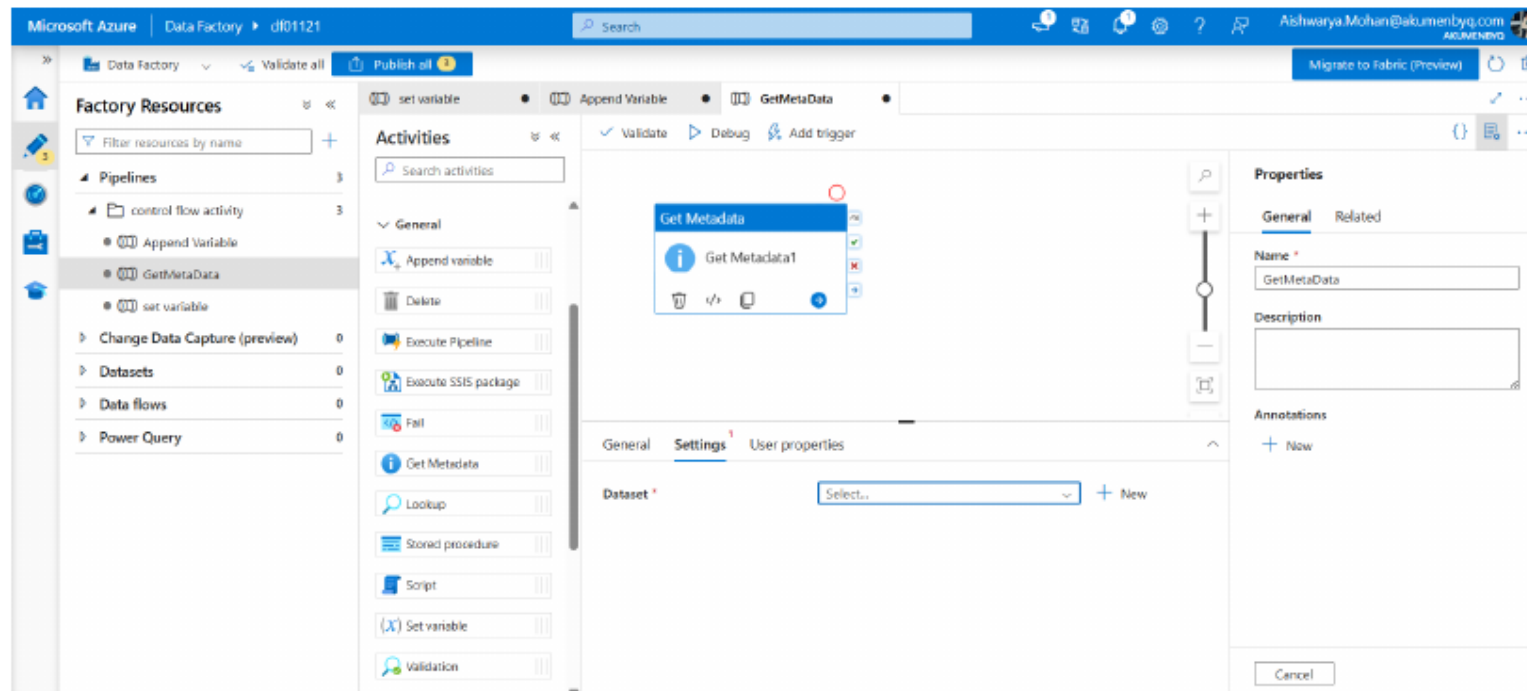
. The 'Properties' pane on the right shows the 'Name' as 'Append Variable'. The bottom status bar indicates that the 'Set variable1' activity has 'Succeeded' at '8/18/2025, 6:26:35 PM'.

Step 12: Get Metadata Activity

Create a new pipeline (GetMetaData).

Add Get Metadata activity and select a dataset (e.g. DelimitedText1).

Choose the fields to retrieve. Pass the output to a Set variable activity to view it.



Step 12 (continued): Get Metadata – Dataset and Field List

Dataset is selected and Field list is configured.

The metadata returned can be inspected via the connected Set variable activity.

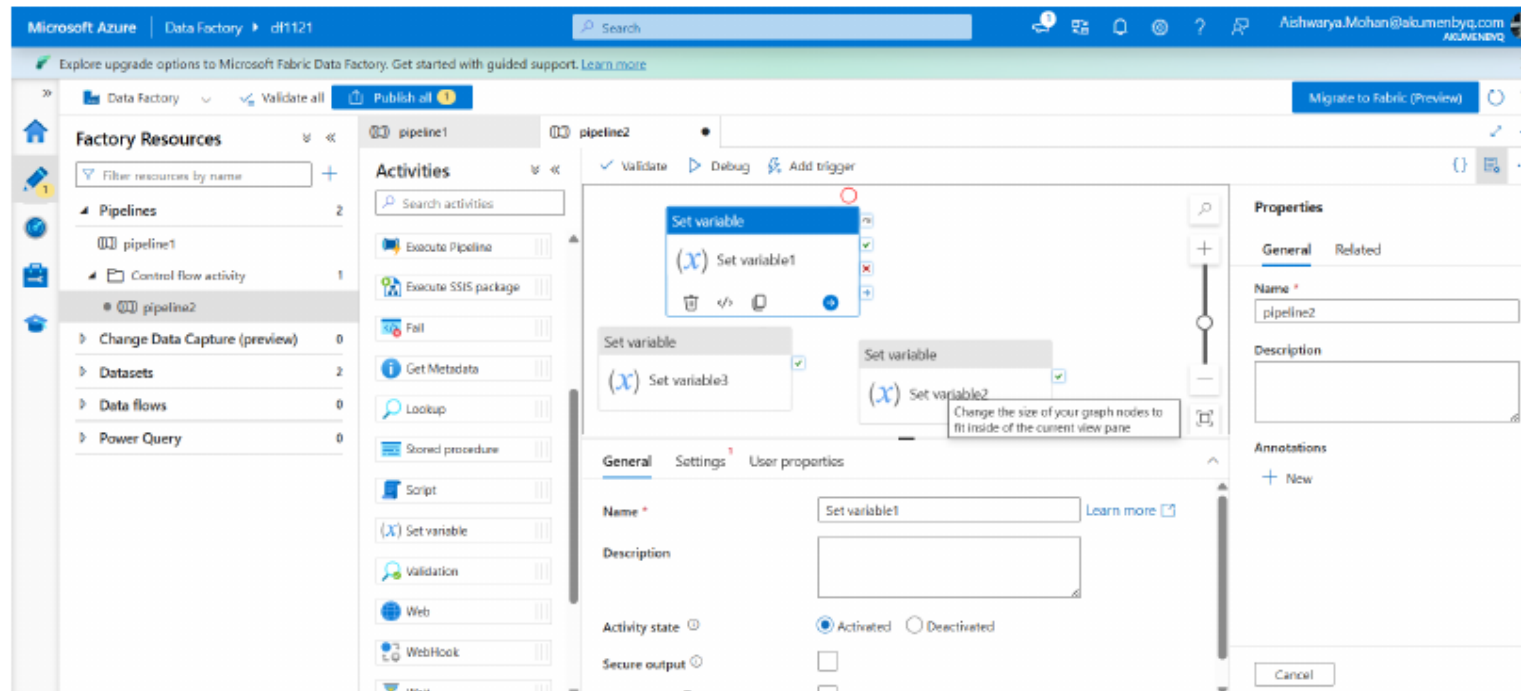
The screenshot displays the Microsoft Azure Data Factory (ADF) interface for a pipeline named 'df01121'. The 'Factory Resources' pane on the left shows the hierarchy: Pipelines > control flow activity > GetMetadataData. The 'Activities' pane in the center lists various activities, with 'Get Metadata' selected. The main canvas shows a 'Get Metadata' activity named 'Get Metadata1' connected to a 'DelimitedText1' dataset. The 'Properties' pane on the right is open, showing the 'General' tab with the activity name 'GetMetadata' and a description field. The 'Settings' tab is also visible, showing the 'Dataset' dropdown set to 'DelimitedText1' and a 'Field list' dropdown set to '+ New'. The 'Filter by last modified' section includes 'Start time (UTC)' and 'End time (UTC)' input fields. A 'Cancel' button is located at the bottom right of the settings pane.

Step 13: Execute Pipeline Activity

Add Execute Pipeline activity to the parent pipeline.

Select the child pipeline to execute and pass any required parameters.

Capture the output of the child pipeline (using Set variable or Pipeline return value)
so the result is visible. Both parent and child pipelines should show Succeeded status.



Summary

All Control Flow activities have been successfully configured and executed:

- Copy Data activity with column mapping
 - Delete activity with logging settings
- Manual, Schedule, Tumbling Window and Storage Event triggers
- Set Variable – area calculation (fixed values + parameterised runtime input)
 - Append Variable
 - Get Metadata
 - Execute Pipeline

This project demonstrates practical use of Azure Data Factory Control Flow activities on the Microsoft Azure platform.

Aishwarya Mohan